

Naive Bayes Classifier | Part 7 | Mathematics Behind the Algorithm

1 Start with Bayes' Theorem

For a class C_k and features $x_1, x_2, \dots, x_n$:

$$P(C_k|x_1, x_2, \dots, x_n) = \frac{P(x_1, x_2, \dots, x_n|C_k) \cdot P(C_k)}{P(x_1, x_2, \dots, x_n)}$$

- $P(C_k) \rightarrow$ Prior probability of class C_k
 - $P(x_1, x_2, \dots, x_n|C_k) \rightarrow$ Likelihood of features given the class
 - Denominator $P(x_1, x_2, \dots, x_n)$ is the same for all classes, so we often ignore it when comparing classes.
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2 The Naive Assumption (Independence)

We assume features are **conditionally independent** given the class:

$$P(x_1, x_2, \dots, x_n|C_k) = \prod_{i=1}^n P(x_i|C_k)$$

👉 This simplifies computation dramatically!

3 The Naive Bayes Formula

So the posterior becomes:

$$P(C_k|x_1, x_2, \dots, x_n) \propto P(C_k) \cdot \prod_{i=1}^n P(x_i|C_k)$$

We pick the class with the maximum posterior probability:

$$\hat{C} = \arg \max_{C_k} P(C_k) \cdot \prod_{i=1}^n P(x_i|C_k)$$

4 Log Transformation (For Numerical Stability)

Multiplying many small probabilities can cause **underflow** (resulting in numbers too small for computers to handle).

So instead, we use **logarithms**:

$$\hat{C} = \arg \max_{C_k} \left(\log P(C_k) + \sum_{i=1}^n \log P(x_i|C_k) \right)$$

👉 Logs convert multiplication into addition → much easier & numerically stable.

5 Example (Text Classification)

Suppose we want to classify whether a message is **Spam** or **Not Spam** based on words:

- Features = {"offer", "click"}
- Formula:

$$P(\text{Spam}|\text{offer}, \text{click}) \propto P(\text{Spam}) \cdot P(\text{offer}|\text{Spam}) \cdot P(\text{click}|\text{Spam})$$

$$P(\text{NotSpam}|\text{offer}, \text{click}) \propto P(\text{NotSpam}) \cdot P(\text{offer}|\text{NotSpam}) \cdot P(\text{click}|\text{NotSpam})$$

We compute both and choose whichever is larger.

6 Key Takeaways

- **Naive Bayes math = Bayes' Theorem + Independence assumption.**
- Final formula:

$$\hat{C} = \arg \max_{C_k} P(C_k) \cdot \prod_{i=1}^n P(x_i|C_k)$$

- In practice, we often use **log form** to avoid underflow.
 - The algorithm is simple, fast, and effective, especially for text data.
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✓ Summary:

The mathematics behind Naive Bayes reduces a complex joint probability into a simple product of conditional probabilities using the **independence assumption**, making classification computationally efficient.
